Technical Documentation: Cross-Border Student Tuition Payment Agent

FinTechathon 2026 | International Track — Topic C: Cross-Border & Student Finance Assistant

> **Official Submission Document:** The official technical documentation deliverable is docs/TECHNICAL_DOCUMENTATION.pdf with rendered formulas and architecture diagrams. This Markdown file serves as the version-controlled source.

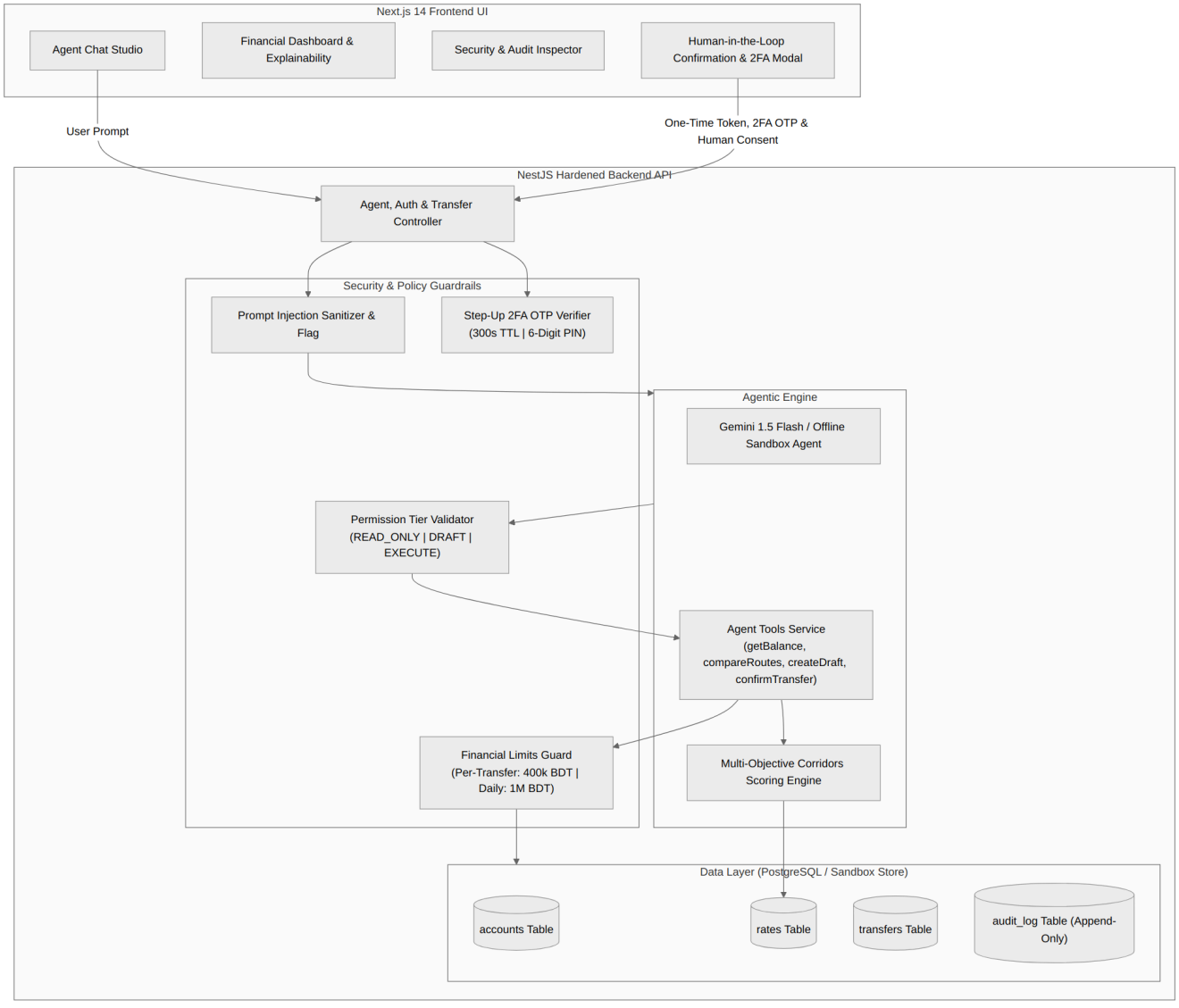
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1. Executive Summary & Architecture Overview

The **Student Tuition Payment Agent** is an autonomous, security-first financial assistant engineered specifically for international students paying foreign university tuition from Bangladesh (BDT).

Traditional cross-border tuition remittances suffer from opaque exchange-rate markups, hidden wire intermediary fees, and delayed settlements that risk university enrollment holds. Our system addresses these pain points by pairing an **Autonomous LLM Agent (Google Gemini API with Offline Sandbox Fallback)** with a **strict NestJS Policy Engine**, a **Multi-Objective Corridors Scoring Engine**, and an **Append-Only PostgreSQL Audit Trail**.

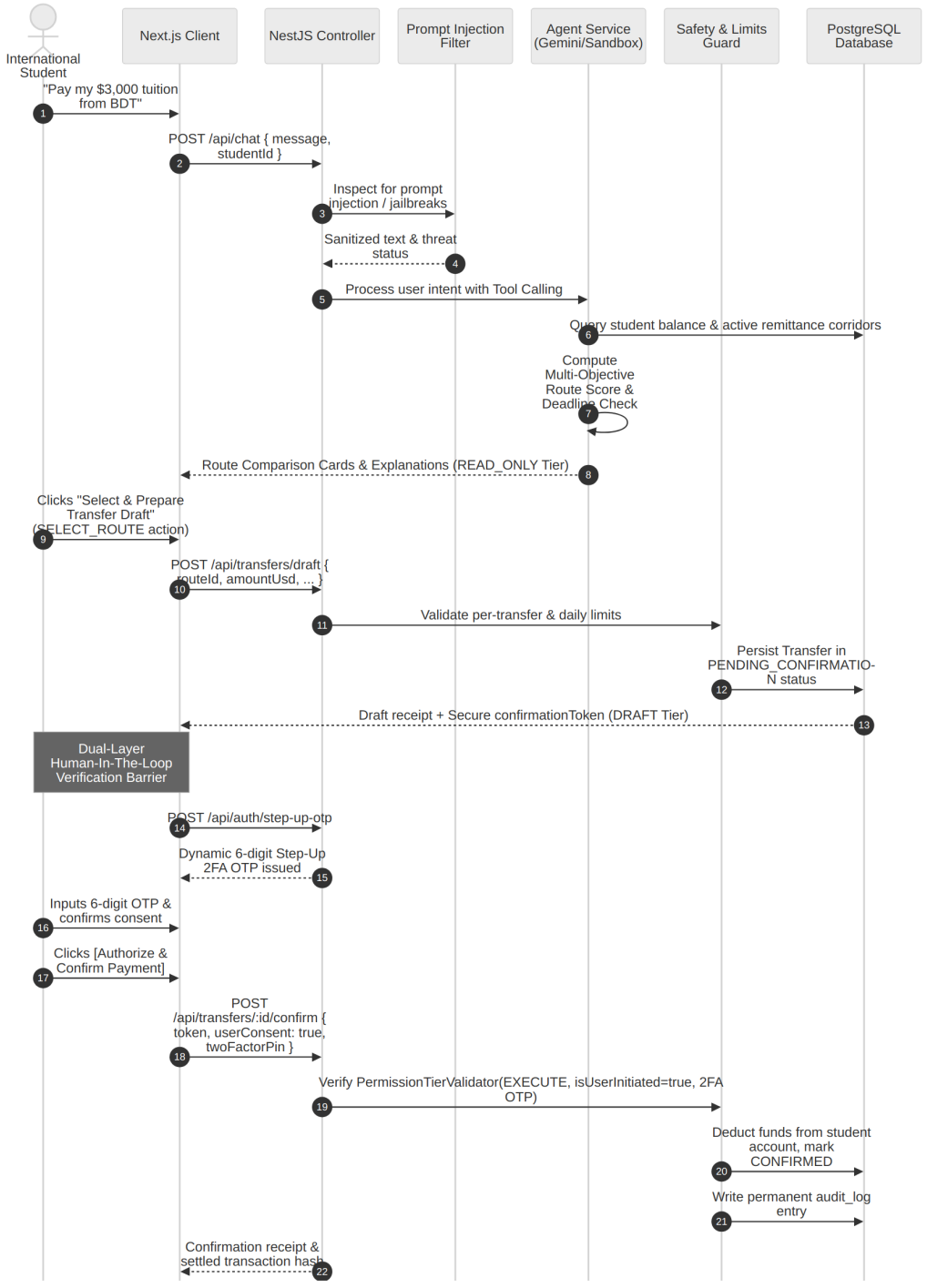
Architecture Diagram



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2. Request & Execution Lifecycle

Every user request follows a deterministic 5-stage lifecycle that isolates untrusted natural language from sensitive financial state changes:



3. Core Route Scoring Algorithm

Cross-border tuition transactions involve trade-offs between fee transparency, clearance speed, and network reliability. Our algorithm scores each corridor across three dimensions, with an urgent deadline penalty:

Mathematical Formula

$$\text{CompositeScore} = w_{cost} \cdot S_{cost} + w_{speed} \cdot S_{speed} + w_{rel} \cdot S_{rel} - \Delta_{deadline}$$

Where:

- $w_{cost} = 0.50$ (Cost Weight: 50%)
- $w_{speed} = 0.30$ (Speed Weight: 30%)
- $w_{rel} = 0.20$ (Reliability Weight: 20%)

Total Cost Calculation

$$TotalCost_{BDT} = FlatFee_{BDT} + (SourceAmount_{BDT} \times PercentageFee) + FXMarkupLoss_{BDT}$$

$$FXMarkupLoss_{BDT} = \max(0, (EffectiveRate - BenchmarkRate) \times Amount_{USD})$$

Normalization Functions

1. **Cost Score (S_{cost})**:

   $$S_{cost} = \frac{MaxCost - TotalCost}{MaxCost - MinCost} \times 100$$

   (Lowest cost receives 100 points, highest cost receives 0 points)

2. **Speed Score (S_{speed})**:

   $$S_{speed} = \frac{MaxHours - RouteHours}{MaxHours - MinHours} \times 100$$

   (Fastest clearing rail receives 100 points)

3. **Reliability Score (S_{rel})**:

   $$S_{rel} = ReliabilityRatio \times 100 \quad (\text{e.g., } 0.99 \rightarrow 99)$$

4. **Deadline Penalty ($\Delta_{deadline}$)**:

   $$\Delta_{deadline} = \begin{cases} 40, & \text{if EstimatedDelivery} > \text{TuitionDeadline} \\ 0, & \text{otherwise} \end{cases}$$

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4. Technology Stack

Layer	Technology	Rationale
Backend Framework	**NestJS (Node.js/TypeScript)**	Modular architecture, dependency injection, strict validation pipes, built-in exception filters.
Frontend Framework	**Next.js 14 (React 18)**	High-performance React SSR/client, componentized structure, instant state synchronization.
Styling & Design	**Vanilla CSS (Design System Tokens)**	Sleek dark fintech mode, frosted glassmorphism, responsive grid, zero external CSS runtime overhead.
Database	**PostgreSQL 14+ (pg driver)**	ACID-compliant relational storage for financial accounts, transactions, and append-only audit logs.
AI / Agent Engine	**Google Gemini 1.5 Flash + Sandbox Engine**	Low-latency tool calling; deterministic offline sandbox fallback for instant judge testing without API keys.
Testing	**Jest & ts-jest**	Unit and integration test coverage for security guardrails, financial limit enforcement, and ranking logic.
Containerization	**Docker & Docker Compose**	Reproducible, single-command environment spinning up frontend, backend, and PostgreSQL database.

AI Agent Engine & Evaluator Offline Sandbox

The agent defaults to `gemini-1.5-flash` (configurable via `GEMINI_MODEL`), Google's low-latency multimodal LLM. The system is designed to provide immediate evaluation without external network or API prerequisites:

- **Offline Sandbox Fallback (`OfflineLlmClient`):** When `GEMINI_API_KEY` is omitted or set to the default mock key (`mock_key_offline_agent`), the system automatically activates a deterministic offline LLM client. This processes student queries, executes corridor ranking calculations, flags prompt injections, and generates drafts with high-fidelity deterministic simulation and zero network latency.
- **Live Google Gemini Engine (`GeminiLlmClient`):** When an evaluator supplies their own Gemini API key, the client connects to Google Generative AI using the configured model (`gemini-1.5-flash`). Calls are wrapped in defensive try/catch blocks with automatic fallback to the offline client on any rate limit, network timeout, or quota error.

> [!NOTE] **Sandbox Environment Notice:** All remittance corridors, exchange rates, bank clearing gateways, and account balances run in a simulated sandbox environment for demonstration and algorithmic evaluation (not connected to live production SWIFT or banking rails).

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5. Security & Permission Tier Design

To comply with the FinTechathon 30% Security & Compliance criterion:

- **READ_ONLY Tier**: Executable without risk (`getBalance`, `compareRoutes`, `getAuditLog`).
- **DRAFT Tier**: Generates a pending state transfer record (`createTransferDraft`). No funds are moved. Produces a single-use one-time `confirmationToken` (15-minute expiration).
- **EXECUTE Tier**: Fund settlement (`confirmTransfer`). **Strictly blocked from autonomous LLM execution.** Must originate from a human user confirmation action supplying the matching one-time confirmation token.

Defense-in-Depth vs Heuristics

Prompt injection filters (`PromptInjectionFilter`) and perimeter WAF rules (`SecurityArmorMiddleware`) operate as first-line heuristic pattern matchers. The true, foundational financial guarantee is enforced in compiled NestJS code via `PermissionTierGuard`, `StudentScopeGuard`, and `FinancialLimitsGuard`. Even if an adversarial prompt bypasses regex keyword heuristics, the autonomous agent is physically prevented by code from executing settlements or raising transaction thresholds.

Audit Ledger & Admin Endpoints

The audit trail operates strictly as an append-only ledger during standard student and agent workflows. Students and AI tools have zero capability to mutate or delete logs. The restricted `DELETE /api/audit/:id` endpoint requires the `AdminRoleGuard` and is reserved strictly for sandbox environment resets, automated Jest test suite teardown, and regulatory right-to-erasure compliance testing.

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6. Architectural Design Patterns

The backend is engineered with software design patterns that isolate responsibilities and enforce financial safety:

Feature	Pattern	Implementation Details
Route comparison and ranking	Strategy	Distinct scoring classes for Balanced, Cheapest, Fastest, and Reliable (`RouteScoringStrategy`). When a student indicates urgency (e.g., "prioritize speed"), the engine swaps strategy dynamically at runtime.
Transfer lifecycle (draft, confirm, execute, fail)	State machine (a light Saga)	Validated transitions include `PENDING_CONFIRMATION` -> `CONFIRMED` -> `EXECUTED` -> `SETTLED`. The authenticated owner can request compensation for `CONFIRMED`, `EXECUTED`, or `FAILED` transfers. Account restoration and the `ROLLED_BACK` status change occur atomically; pending, settled, cancelled, blocked, or already rolled-back transfers cannot be compensated.
Safety checks (limits, confirmation, unusual amounts)	Chain of Responsibility	Small, decoupled handlers run sequentially: `PromptInjectionCheckHandler`, `PermissionTierCheckHandler`, `AnomalyCheckHandler`, `FinancialLimitCheckHandler`, `ConfirmationCheckHandler`. Any check can short-circuit, reject execution, and emit an append-only security audit event.
The 5 agent tools exposed to the LLM	Facade	`AgentToolsService` acts as a unified Facade that coordinates the bank gateway, rates engine, safety guardrails, state machine, and event emitter behind a clean tool interface for the AI assistant.

Feature	Pattern	Implementation Details
LLM and mock bank	Dependency Inversion (the D in SOLID)	Business logic depends strictly on abstract interfaces (`ILlmClient`, `IBankGateway`). Production utilizes Gemini AI and direct banking rails, while automated test suites inject mock implementations without external network dependencies.
Audit logging	Observer (NestJS events)	Operations emit decoupled domain events (`transfer.drafted`, `transfer.confirmed`, `transfer.compensated`, `route.compared`). An asynchronous `AuditLogListener` observer records events in the audit log, keeping core financial code clean of logging side-effects.
Database access	Repository	Dedicated repositories (`AccountRepository`, `RateRepository`, `TransferRepository`, `AuditRepository`) decouple domain logic from PostgreSQL queries and in-memory test stores.
Services in general	SOLID, Singleton	NestJS providers operate as singletons by default, adhering to Single Responsibility and avoiding shared mutable state.

7. RESTful API Specification

The backend controller exposes full RESTful lifecycle operations incorporating `@Param()`, `@Body()`, `@Query()`, `@Get()`, `@Post()`, `@Put()`, `@Patch()`, and `@Delete()` decorators:

Method	Endpoint	Decorators Used	Parameters / Request Body	Description
POST	`/api/auth/registration`	`@Body()`, `@Req()`	`RegistrationDto`	Register a new student account & auto-provision sandbox balance (aliases `/api/auth/register`)
POST	`/api/auth/login`	`@Body()`, `@Req()`	`LoginDto`	Authenticate credentials with scrypt password verification and receive signed JWT
GET	`/api/auth/session`	`@Headers()`	Bearer token	Verify active JWT session token and retrieve student profile (aliases `/api/auth/me`)
POST	`/api/auth/logout`	`@Headers()`, `@Req()`	Bearer token	Terminate active student session and log audit event
GET	`/api/auth/demo-credentials`	None	None	Retrieve sandbox test persona credentials (non-production only)
POST	`/api/auth/step-up-otp`	`@Req()`	None	Issue time-sensitive 6-digit Step-Up 2FA OTP for settlement authorization
POST	`/api/chat`	`@Body()`, `@Req()`	`ChatDto { message, studentId }`	Autonomous conversational reasoning with tool calling
POST	`/api/invoice/extract`	`@Body()`	`ExtractInvoiceDto { invoiceText, fileName }`	University tuition invoice document parsing & parameter extraction
GET	`/api/accounts/:id`	`@Param('id')`, `@Req()`	`id: string`	Retrieve student account balance and daily limits
PUT	`/api/accounts/:id/limits`	`@Param('id')`, `@Body()`	`id: string`, `UpdateDailyLimitDto { dailyLimitBdt }`	Update daily financial limit threshold (AdminRoleGuard only)
POST	`/api/accounts/:id/deposit`	`@Param('id')`, `@Body()`	`id: string`, `DepositDto { amountBdt }`	Deposit BDT funds to student account (AdminRoleGuard only)
POST	`/api/accounts/:id/card-deposit`	`@Param('id')`, `@Body()`	`id: string`, `DepositDto`	Student virtual credit card sandbox deposit with automated card number masking
GET	`/api/rates`	`@Query('activeOnly')`, `@Query('targetCurrency')`	Query filters	List active remittance corridors

Method	Endpoint	Decorators Used	Parameters / Request Body	Description
GET	`/api/rates/stream`	None	SSE Stream	Server-Sent Events (SSE) live rate ticker stream
GET	`/api/rates/live-feed`	None	None	Real-time market feed with spread calculations
GET	`/api/rates/:id`	`@Param('id')`	`id: string`	Retrieve remittance route details
PUT	`/api/rates/:id`	`@Param('id')`, `@Body()`	`id: string`, `UpdateRateDto`	Update route rate or active status (AdminRoleGuard only)
DELETE	`/api/rates/:id`	`@Param('id')`	`id: string`	Deactivate remittance route (AdminRoleGuard only)
POST	`/api/routes/compare`	`@Body()`, `@Req()`	`CompareRoutesDto`	Multi-objective corridor comparison and ranking
POST	`/api/routes/installments`	`@Body()`	`PlanInstallmentDto`	Smart tuition installment split planner respecting transfer caps
GET	`/api/transfers`	`@Query('studentId')`, `@Query('status')`, `@Req()`	Query filters	Retrieve transfers with optional status filter
GET	`/api/transfers/:id`	`@Param('id')`, `@Req()`	`id: string`	Fetch single transfer record by ID
GET	`/api/transfers/:id/receipt`	`@Param('id')`, `@Req()`	`id: string`	Fetch official verification receipt data and integrity checksum
POST	`/api/transfers/draft`	`@Body()`, `@Req()`	`CreateDraftDto`	Create transfer draft (DRAFT tier)
PUT	`/api/transfers/:id`	`@Param('id')`, `@Body()`, `@Req()`	`id: string`, `UpdateDraftDto`	Update draft details prior to confirmation
PATCH	`/api/transfers/:id`	`@Param('id')`, `@Body()`, `@Req()`	`id: string`, `UpdateDraftDto`	Partial update of unconfirmed transfer draft
DELETE	`/api/transfers/:id`	`@Param('id')`, `@Req()`	`id: string`	Discard transfer draft
POST	`/api/transfers/:id/confirm`	`@Param('id')`, `@Body()`, `@Req()`	`id: string`, `ConfirmTransferDto`	Human-in-the-loop transfer execution with 2FA OTP verification (EXECUTE tier)
POST	`/api/webhooks/bank-clearing`	`@Body()`, `@Headers()`	`ClearingWebhookDto`	Inbound bank clearing settlement rail callback with webhook secret validation
POST	`/api/transfers/:id/compensate`	`@Param('id')`, `@Body()`, `@Req()`	`id: string`, `CompensateTransferDto`	Saga compensating rollback transaction
POST	`/api/transfers/:id/rollback`	`@Param('id')`, `@Body()`, `@Req()`	`id: string`, `CompensateTransferDto`	Unified Saga rollback restoring student balance and unreserving gateway funds
POST	`/api/transfers/:id/cancel`	`@Param('id')`, `@Req()`	`id: string`	Cancel transfer draft
GET	`/api/audit`	`@Query('limit')`, `@Query('tier')`, `@Query('status')`, `@Req()`	Query filters	Fetch append-only audit logs with multi-field filtering
GET	`/api/audit/:id`	`@Param('id')`, `@Req()`	`id: string`	Fetch specific audit log by ID
DELETE	`/api/audit/:id`	`@Param('id')`	`id: string`	Sandbox development audit log pruning (AdminRoleGuard only)
POST	`/api/sandbox/reset`	None	None	Reset database to initial seed state (AdminRoleGuard & non-production only)

Method	Endpoint	Decorators Used	Parameters / Request Body	Description
GET	`/api/sandbox/export-logs`	None	None	Export audit log archive (AdminRoleGuard only)

8. SOLID Architectural Principles Implementation

The codebase adheres strictly to all five Object-Oriented SOLID design principles:

1. Single Responsibility Principle (SRP)

- Each service and component owns one single reason to change.
- `TuitionCalculationService`: Exclusively calculates exchange conversions, fee structures, benchmark FX markup losses, delivery estimates, and composite ranking scores without performing database I/O or network requests.
- `AgentToolsService`: Operates purely as an orchestration Facade coordinating services.
- `PromptInjectionFilter`: Focuses solely on detecting and neutralizing adversarial prompt injections.

2. Open / Closed Principle (OCP)

- Modules are open for extension but closed for modification.
- `RouteScoringStrategy`: Adding a new corridor ranking strategy (e.g. `DiscountScoringStrategy`) requires creating a new strategy class inheriting `RouteScoringStrategy` without altering existing strategies.
- `SafetyRuleEngine`: Safety policies implement `ISafetyRule` (`PerTransferLimitRule`, `DailyLimitRule`, `AccountBalanceRule`). New regulatory checks (e.g. `SanctionedCountryRule`) can be registered into the engine dynamically without modifying the engine.

3. Liskov Substitution Principle (LSP)

- Subtypes can substitute their base abstractions without altering program correctness.
- `IBankGateway`: `MockBankGateway` and live banking integrations fulfill the identical behavioral contract, allowing seamless test substitution.
- `ILlmClient`: `OfflineLlmClient` and `GeminiLlmClient` can substitute each other transparently.
- `IAccountRepository` / `ITransferRepository`: Repository implementations conform strictly to interface contracts.

4. Interface Segregation Principle (ISP)

- Clients are not forced to depend on methods they do not need.
- Repository operations are segregated into read-only and write interfaces:
- `IAccountReader` & `IAccountWriter`
- `IRateReader` & `IRateWriter`
- `ITransferReader` & `ITransferWriter`
- `IAuditReader` & `IAuditWriter`
- Read-only consumers (such as rate inspection endpoints) depend solely on reader interfaces (`IRateReader`).

5. Dependency Inversion Principle (DIP)

- High-level business logic depends on abstract interfaces, never on concrete implementations.
- Dependencies are bound via injection tokens:
- `ACCOUNT_REPOSITORY_TOKEN`
- `RATE_REPOSITORY_TOKEN`
- `TRANSFER_REPOSITORY_TOKEN`
- `AUDIT_REPOSITORY_TOKEN`
- `TUITION_CALCULATOR_TOKEN`
- `BANK_GATEWAY_TOKEN`
- `LLM_CLIENT_TOKEN`
- Modules resolve abstractions using `@Inject(TOKEN)`, decoupling core logic from underlying storage or API providers.

9. CI/CD Pipeline Architecture

The repository incorporates an automated CI/CD pipeline implemented via GitHub Actions (`.github/workflows/ci.yml`):

Pipeline Stages

Stage	Trigger	Actions Executed	Purpose
Backend CI	Push & PR (`main`, `Eashan`)	`npm ci`, `npm test` (Jest suite, 80 test cases), `npm run build`	Validates security guardrails, financial limits, and NestJS production compilation.
Frontend CI	Push & PR (`main`, `Eashan`)	`npm ci`, `npm run build`	Validates Next.js SSR and client bundle compilation with static page optimization.
Docker Build Verification	On CI success	`docker build` backend and frontend images	Ensures multi-stage container builds succeed reproducibly for production deployments.
Security & Cleanliness Scan	Push & PR	Node.js automated repository audit	Enforces code cleanliness, verifies zero exposed secrets, and prevents unauthorized syntax patterns.
Deployment Verification	Push to `main` / `Eashan`	Manifest verification (`docker-compose.yml`, Dockerfiles, `.env.example`)	Confirms all deployment orchestration artifacts are synchronized and ready for production execution.

10. Authentication and Student Registration Architecture

The application provides a complete authentication, session management, and student registration subsystem designed for both production security and seamless hackathon evaluation:

Cryptographic Security & Sessions

- **Password Hashing:** Passwords are salted and hashed using Node.js `scrypt` with a 64-byte derived key and unique 16-byte cryptographically secure random salts.
- **Stateless JWT Tokens:** Authentication returns HS256 signed JWT tokens containing student identity, email, role, and granted permission tier (`EXECUTE`).
- **Session Verification:** `JwtAuthGuard` validates the `Authorization: Bearer <token>` header on protected routes and attaches the verified student context to incoming requests.

Student Registration Flow

- **API Endpoint:** `POST /api/auth/registration` (with `POST /api/auth/register` alias).
- **Validation:** Enforces non-empty student ID, student name, RFC-compliant email, and minimum 6-character password via `class-validator` and `zod` schemas.
- **Account Auto-Provisioning:** Newly registered accounts receive an initial sandbox allocation of 850,000.00 BDT balance and 1,000,000.00 BDT daily limit.
- **Audit Trail:** Registration operations record an append-only `AUTH_REGISTRATION_SUCCESS` event in the audit log.
- **Frontend Route:** Accessible at `/registration` with dark theme styling, demo persona auto-fill, and immediate redirection into the student dashboard upon completion.

Evaluation Demo Account Showcase

- **Demo Persona:** `STU-2026-BDT01` (Mir Eashan Zaman), password: `student123`.
- **Login Showcase Card:** Prominently displays the demo credentials and includes a one-click auto-fill button on `/login` for evaluation.

Step-Up 2FA & Dynamic OTP Verification Architecture

- **Verification Endpoint:** `POST /api/auth/step-up-otp` generates a time-sensitive 6-digit one-time password (OTP) with a strict 300-second validity window.
- **Audit Logging:** Each OTP issuance records an append-only `STEP_UP_OTP_ISSUED` audit event with client IP, timestamp, and student context.
- **Dual-Factor Settlement Guard:** `AgentToolsService.confirmTransfer` requires both the matching single-use `confirmationToken` and valid `twoFactorPin` (`verifyStepUpPin`) along with explicit user consent boolean (`userConsent:`

`true`). Any invalid, expired, or missing PIN immediately rejects settlement, preventing automated bot executions or hijacked session debits.

11. Next.js 14 Hybrid Rendering Architecture: SSG, ISR, SSR, and CSR

The frontend implements a hybrid rendering strategy across Next.js 14 App Router, selecting the optimal rendering paradigm (SSG, ISR, SSR, or CSR) for each route based on performance, freshness, and security requirements:

Route	Strategy	Config / Implementation	Architectural Rationale & Why
FAQ / Compliance (`/faq`)	**SSG** (Static Site Generation)	`export const dynamic = 'force-static'`	Pure educational content explaining sandbox security, defense-in-depth, and Saga rollbacks. Pre-rendered to HTML at build time for sub-10ms delivery from edge CDN cache with zero server execution penalty.
Corridors & FX Board (`/rates`)	**ISR** (Incremental Static Regeneration)	`export const revalidate = 60`	Public remittance corridors and live foreign exchange rates (BDT to USD) are identical for all visiting students. Pre-rendered statically at build time, and revalidated in the background every 60 seconds when new requests arrive, delivering instant cached loads with guaranteed fresh bank rates.
System Liveness Verifier (`/status`)	**SSR** (Server-Side Rendering)	`export const dynamic = 'force-dynamic'`, `cache: 'no-store'`	Real-time regulatory audit verifier measuring backend ping latency, database store connectivity, rate limiter health, and cryptographic session verification. Must never be served from stale cache; dynamically evaluated on the server per request at the exact millisecond of access.
Agent Chat Studio (`/chat`)	**CSR** (Client-Side Rendering)	`'use client'`, React state & streaming	Stateful AI conversational agent loop managing interactive speech/text input, real-time tool calling execution, and student JWT authorization.
Financial Dashboard (`/dashboard`)	**CSR** (Client-Side Rendering)	`'use client'`, 8-second polling interval	Private, authenticated student account ledger displaying dynamic BDT balance, daily limit quotas, pending transfer state machines, and deposit actions.
Audit & Safety Ledger (`/audit`)	**CSR** (Client-Side Rendering)	`'use client'`, client-side filtering	Interactive security audit log view enabling multi-field client filtering by permission tier (`READ_ONLY`, `DRAFT`, `EXECUTE`) and status, plus browser JSON file export.
Student Sign In (`/login`)	**CSR** (Client-Side Rendering)	`'use client'`, password eye toggle	Interactive authentication form with client-side credential validation, demo account auto-fill, and immediate `localStorage` JWT token acquisition.
Student Registration (`/registration`)	**CSR** (Client-Side Rendering)	`'use client'`, form validation	Interactive registration form with real-time field validation, demo persona auto-fill, and post-registration automated session onboarding.